15.081 Mathematical programming, HW6

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1. **(Theorem of alternative)** Given matrices A, B, C, D and vectors e, f, consider the linear system

$$Ax + By \geq e \qquad Cx + Dy = f \qquad x \geq 0 \qquad y \text{ free}$$

(Want) State and prove a Theorem of Alternative for such a system.

Theorem of alternative: Exactly one holds

$$\begin{aligned}(a)\ \exists\ x \geq 0,\ y: &\quad Ax + By \geq e,\ Cx + Dy = f \\ (b)\ \exists\ p \geq 0,\ q: &\quad p^\top B + q^\top D = 0,\ p^\top A + q^\top C \leq 0,\ p^\top e + q^\top f > 0\end{aligned}$$

Proof: **($(a) \Rightarrow \neg(b)$)** Contradiction (assume (a) and (b))

$$\begin{aligned} p^\top(Ax + By) &\geq p^\top e && (\forall\ p \geq 0, \text{ since } Ax + By \geq e \text{ from } (a)) \\ p^\top(Ax + By) + q^\top(Cx + Dy) &\geq p^\top e + q^\top f && (\forall\ q, \text{ since } Cx + Dy = f \text{ from } (a)) \\ (p^\top B + q^\top D)y + (p^\top A + q^\top C)x &> 0 && (\text{since } p^\top e + q^\top f > 0 \text{ from } (b)) \\ (p^\top A + q^\top C)x &> 0 && (\text{since } p^\top B + q^\top D = 0 \text{ from } (b)) \\ p^\top A + q^\top C &\not\leq 0 && (\text{since } x \geq 0 \text{ from } (a)) \end{aligned}$$

Arriving at our contradiction, $p^\top A + q^\top C \not\leq 0$, since we assume $p^\top A + q^\top C \leq 0$. Therefore $(a) \Rightarrow \neg(b)$.

Proof: **($\neg(a) \Rightarrow (b)$)** Consider the primal-dual pair

$$\begin{aligned}(P):&\ \min\ x^\top 0 + y^\top 0 && \text{s.t. } Ax + By \geq e,\ Cx + Dy = f,\ x \geq 0,\ y \text{ free} \\ (D):&\ \max\ p^\top e + q^\top f && \text{s.t. } p \geq 0,\ q \text{ free},\ p^\top A + q^\top C \leq 0,\ p^\top B + q^\top D = 0\end{aligned}$$

(P) infeasible $\Longrightarrow$ (D) unbounded, since $(p, q) = (0, 0)$ is a feasible solution to (D).

(D) unbounded $\Longrightarrow$ $\exists$ feasible (p, q) such that $p^\top e + q^\top f \geq 1 \Longrightarrow p^\top e + q^\top f > 0$.

Feasible $(p, q) \Longrightarrow p \geq 0,\ p^\top A + q^\top C \leq 0,\ p^\top B + q^\top D = 0$. Therefore $\neg(a) \Rightarrow (b)$

2. **(Extreme point of concave optimization)** Consider the problem of the form $\min_{\{x \in P\}} f(x)$ where $f : \mathbb{R}^n \to \mathbb{R}$ is a concave function and P is a polyhedron that does not contain a line. **(Prove)** if the problem has optimal solution(s), then $\exists$ an extreme point of P that is an optimal solution.

(Proof) Assume the problem has optimal solution(s). Pick some optimal $x^{(0)}$. If $x^{(0)}$ is an extreme point, we are done. Now suppose $x^{(0)}$ is not an extreme point. Then

$$\exists \lambda \in (0, 1),\ y, z \in P,\ y \neq z: \quad x^{(0)} = \lambda y + (1 - \lambda) z$$

Concave $f(x)$ gives us

$$f(x^{(0)}) = f(\lambda y + (1-\lambda)z) \geq \lambda f(y) + (1-\lambda)f(z)$$

But $f(x^{(0)})$ optimal means

$$f(x^{(0)}) \leq f(y),\ f(x^{(0)}) \leq f(z) \Longrightarrow f(x^{(0)}) \leq \lambda f(y) + (1-\lambda)f(z)$$

So for any optimal $x^{(0)}$ that is not an extreme point, we must have

$$f(x^{(0)}) = \lambda f(y) + (1-\lambda)f(z) = f(y) = f(z)$$

So y, z are also optimal. If either y or z is an extreme point of P, we are done. Otherwise, both y and z are optimal and non-extreme, so we may apply the above argument to each of them: every optimal point that is not an extreme point can be written as a convex combination of two distinct optimal points in P.

With this process, we obtain a sequence of optimal points, $x^{(0)},\ x^{(1)},\ x^{(2)},\ \ldots$

Since P does not contain a line, this process cannot continue indefinitely. The series is forced to the boundary of P and the process must eventually reach point $x^{(E)}$ where no convex combination is possible. But by definition, $x^{(E)}$ is an extreme point and we are done.

3. **(Linear program reformulation)** Consider the following nonconvex program

$$\min_x \sum_{j=1}^{k} \frac{s_j^\top x + r_j}{p^\top x + q} \qquad \text{s.t. } Ax \geq b$$

Where $s_j, p \in \mathbb{R}^n,\ A \in \mathbb{R}^{m\times n},\ b \in \mathbb{R}^m,\ r_j, q \in \mathbb{R}$. Suppose $p^\top x + q > 0\ \forall\ x : Ax \geq b$. **(Provide)** A linear program that is equivalent to the nonconvex problem.

Hint: consider the change-of-variables: $y = \frac{1}{p^\top x+q}$ *and* $z = yx$. Following the hint, we reformulate the nonconvex problem using change-of-variables in y and then z:

$$\min_{x,y}\ y\sum_{j=1}^{k} s_j^\top x + r_j \qquad \text{s.t. } Ax \geq b,\ y = \frac{1}{p^\top x + q} \qquad (y := \tfrac{1}{p^\top x+q})$$

$$\min_{x,y}\ \sum_{j=1}^{k} s_j^\top (yx) + yr_j \qquad \text{s.t. } A(yx) \geq yb,\ p^\top (yx) + yq = 1,\ y \geq 0$$

(rearranging, multiplying by y, and establishing $y > 0$ from given)

$$\min_{y,z}\ \sum_{j=1}^{k} s_j^\top z + yr_j \qquad \text{s.t. } Az \geq yb,\ p^\top z + yq = 1,\ y \geq 0 \qquad (z := yx)$$

The final reformulation is a linear program: it has a sum of linear terms in the objective ($\sum_{j=1}^{k} s_j^\top z + yr_j$), a set of $m+1$ linear inequality constraints ($Az \geq yb,\ y \geq 0$), and a single linear equality constraint ($p^\top z + yq = 1$).